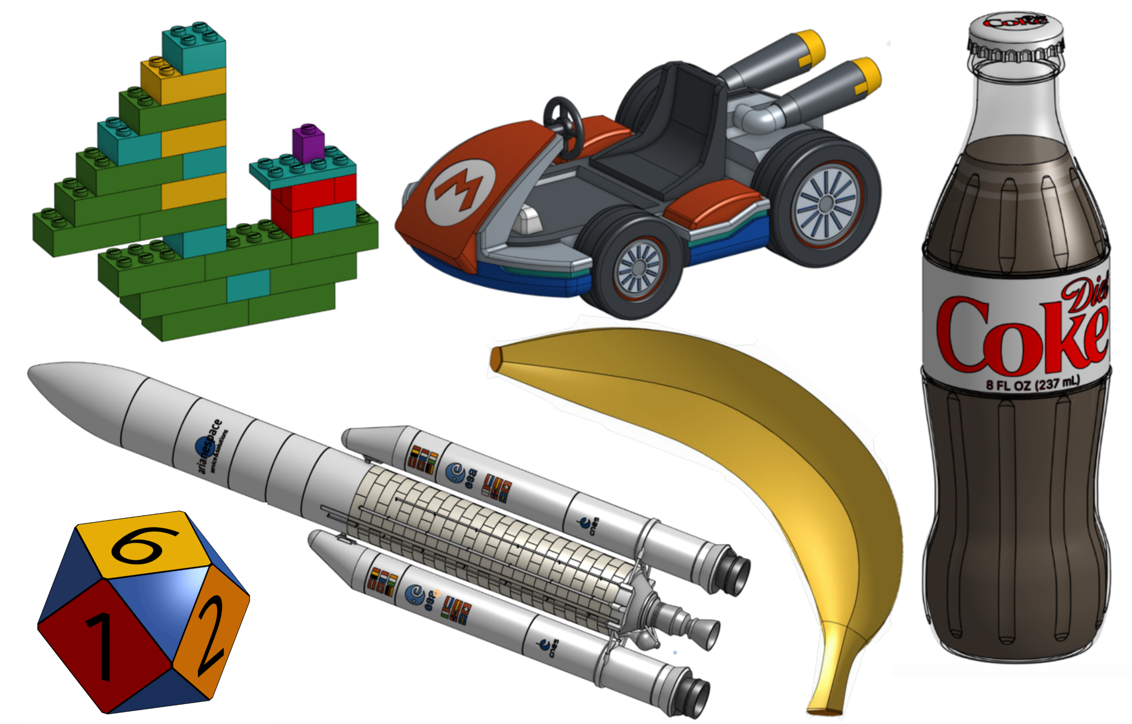
How to CAD Almost Anything! – Disney Edition

Summer 2025 – MIT AeroAstro Workshop

A companion course to the original "How to CAD almost anything!" IAP 2024 workshop, this short, 3-week edition introduces students to the parametric design software Dassault Systèmes SolidWorks using examples sourced from the Disney universe! Although intended for students with existing CAD (computer-aided design) experience, both beginners (no experience at all) and pro-users alike are welcomed! Come learn how to CAD (computer-aided design) almost anything (Disney themed)!



Yes, this could be YOU at the end of the workshop! You'll be equipped with the tools to design cool looking things such as a LEGO boat, a Mario Kart, a bottle of Coke, a banana, an Arianne V rocket, and even a 12-sided dice! These are all projects from "How to CAD Almost Anything! – Summer 2024".

Workshop Details

Subject Title:	How to CAD Almost Anything!
Prerequisites:	Willingness to have fun and think outside the box!
Enrollment:	20.
Attendance:	Participants must attend all sessions.
Meeting Room:	GIS & Data Lab (located on the first floor of the Rotch Library, 7-238).
Meeting Times:	Tuesdays and Thursdays 3pm – 5pm (although may run longer). There is a total of 6 weekly sessions, starting on 06/03/25 until 06/19/25. The sessions will take place on 06/03, 06/05, 06/10, 06/12, 06/17 and 06/20 (a Friday, due to Juneteenth).
Instructor:	Andy Eskenazi - AeroAstro PhD student (LAE), andyeske@mit.edu.

About: Hailing from the capital of Tango, Steak and Football, Andy is currently a PhD student at MIT AeroAstro trying to make aviation more sustainable. Outside of research, he is passionate about all things mechanical design. At MIT, he has taught the "How to CAD Almost Anything!" series four times, on SolidWorks, Fusion, Onshape and NX.

Workshop Description

Ever wondered how are objects from our daily lives designed? How can we generate a computer 3D model of a Sorcerer Mickey Mug, the Magic Kingdom Railroad, or EPCOT's Spaceship Earth? What about designing the Genie's Lamp? The Luxo, Jr. Lamp? Or the Scream Canisters from Monsters, Inc.? In this fun MIT AeroAstro workshop, you will learn the skills to design all of these, and much more!

Split into 6 2-hour long sessions, this course acts as a companion edition to the "How to CAD Almost Anything!" IAP 2024 workshop on Dassault Systèmes SolidWorks. The first two sessions will be focused on reviewing SolidWorks skills, primarily intended for students that did not attend the IAP 2024 workshop. The latter four sessions will be focused purely on modeling, working together to produce advanced Disney-themed CAD models. In contrast to traditional mechanical design courses, this workshop places greater emphasis on the design process itself, understanding how we can plan and best leverage our available tools to arrive to our desired result. Thus, the sessions are less about following the instructions on an engineering drawing, but about independent thinking and strategizing, reverse engineering an object into a 3D model.

Workshop Schedule

#	Month	Day	Date	Outline and Objectives
1	Jun	T	06/03	**Session 1**: Introduction to the SolidWorks – Disney Universe. Objective: In this session, we'll get ourselves acquainted with the SolidWorks workspace and start learning some of the most used tools. Session 1's goals include: • Creating sketches (using basic shapes, construction lines, smart-dimensioning, sketch relationships) and understanding planes. • Understanding what it means for a sketch to be fully defined. • Locating and using the different elementary feature commands (boss extrude, boss cut, fillet, chamfer). • Editing sketches and features after creating them. • Coloring parts and changing material properties. • Learning how to use the spline tool and the wrap command. • Learning how to add a picture and sketch on it. Session activity: Using the tools learned on Session 1, we'll design two objects, namely: • A Sorcerer Mickey mug. • A commemorative Mickey coin for Disneyland Paris.
2	Jun	Th	06/05	**Session 2**: Pixar! Objective: In this session, we'll continue revising some of the most powerful SolidWorks tools. Session 2's goals include: • Understanding how to create a sketch for a revolve.

				• Learning how to make use of the mirroring and circular patterns tools, both as a sketch and as a feature. • Learning how to create planes, at different angles. • Learning how to make an assembly of multiple parts. • Learning how to make an exploded view of an assembly and subsequentially animating it. • Learning how to create an engineering drawing of a part and assembly (including exploded views). <u>Session activities</u>: Using the tools learned on Session 2, we'll design two objects, namely: • A Luxo, Jr. lamp. • A Monsters, Inc. Scream Canister.
3	Jun	T	06/10	**Session 3**: Disney Trains - EPCOT! <u>Objective</u>: In this session, we'll practice using the loft and sweep commands to design to design curved objects. <u>Session activities</u>: We'll design two objects, namely: • EPCOT's Toy Monorail. • EPCOT's Toy Train Tracks.
4	Jun	Th	06/12	**Session 4**: Disney Trains - Magic Kingdom! <u>Objective</u>: In this session, we'll direct our attention towards Disney's iconic locomotive, trying to create an accurate 3D model representation. <u>Session activities</u>: We'll design two objects, namely: • Magic Kingdom's Toy Locomotive. • Magic Kingdom's Toy Wagons. • Magic Kingdom's Toy Train Tracks.
4.5	Jun	Sat	06/14	**Session 4.5 (online-only)**: Disney Trains - Stations. <u>Objective</u>: In this session, we'll continue building the SolidWorks – Disney Universe by practicing how to accurately model a historic-looking building. <u>Session activities</u>: We'll design two objects, namely: • Combined Train Station. • Hybrid Magic Kingdom & EPCOT park layout to place the models from S1 through S6.
5	Jun	T	06/17	**Session 5**: Mickey and the Genie! <u>Objective</u>: In this session, we'll practice using the loft command to design organic-looking geometries. <u>Session activities</u>: We'll design two objects, namely: • The Genie's lamp. • A Mickey statue.

6	Jun	Fr	06/20	**Session 6**: Tomorrowland! Objective: In this session, we'll practice using two advanced modelling techniques, surface modeling and gear mates, to design a geodesic polyhedron and a carousel. Session activities: We'll design two objects, namely: • EPCOT's Spaceship Earth. • Magic Kingdom's Astro Orbiter. • Placing the S1 through S6 models in the park layout.